

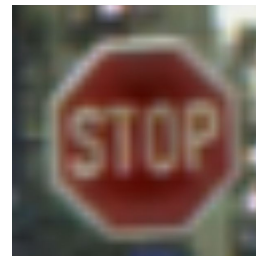
Part I

Security in machine learning

Outline of this lecture

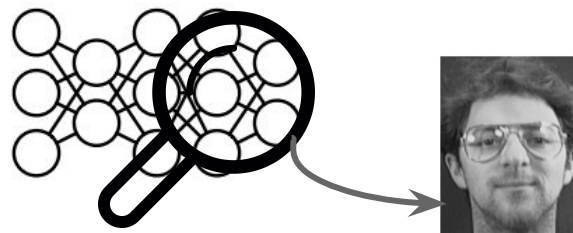
1

Security in ML



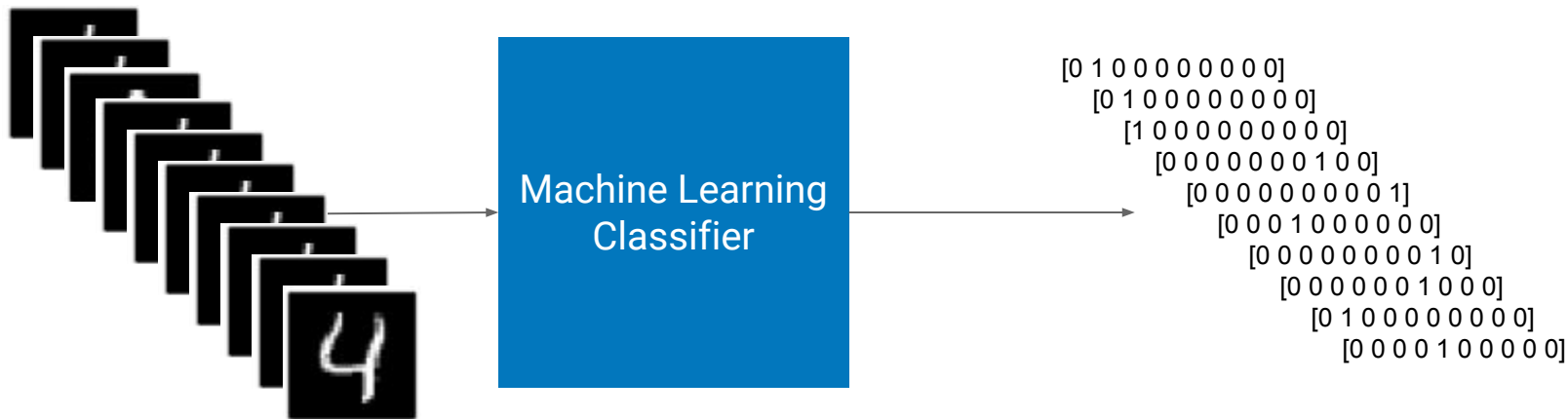
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Privacy in ML





Classifier: map inputs to one class among a predefined set



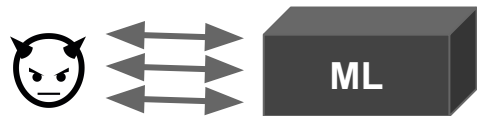
Learning: find internal classifier parameters θ that minimize a cost/loss function ($\sim$ model error)

Attack Models

Attacker may see the model: bad even if an attacker needs to know details of the machine learning model to do an attack --- aka a **white-box attacker**



Attacker may not need the model: worse if attacker who knows very little (e.g. only gets to ask a few questions) can do an attack --- aka a **black-box attacker**



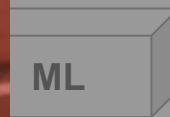
Attack Models

Attacker may see the details of the machine learning model to do a

Attacker may not need to ask a few questions) c



details of the machine

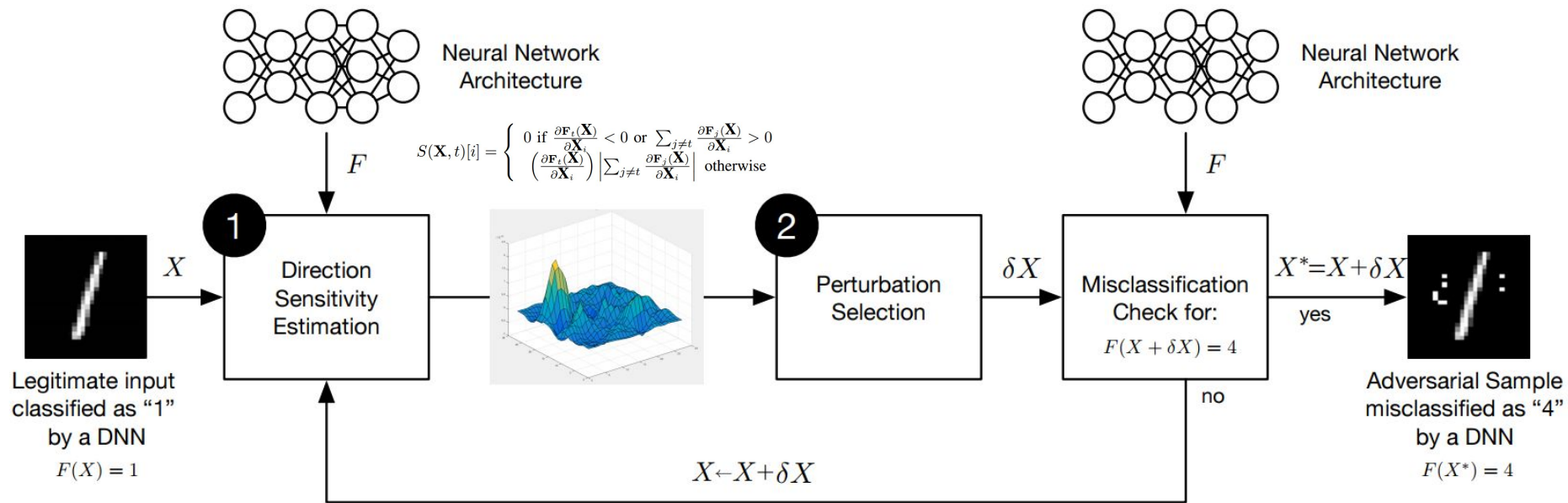


little (e.g. only gets to

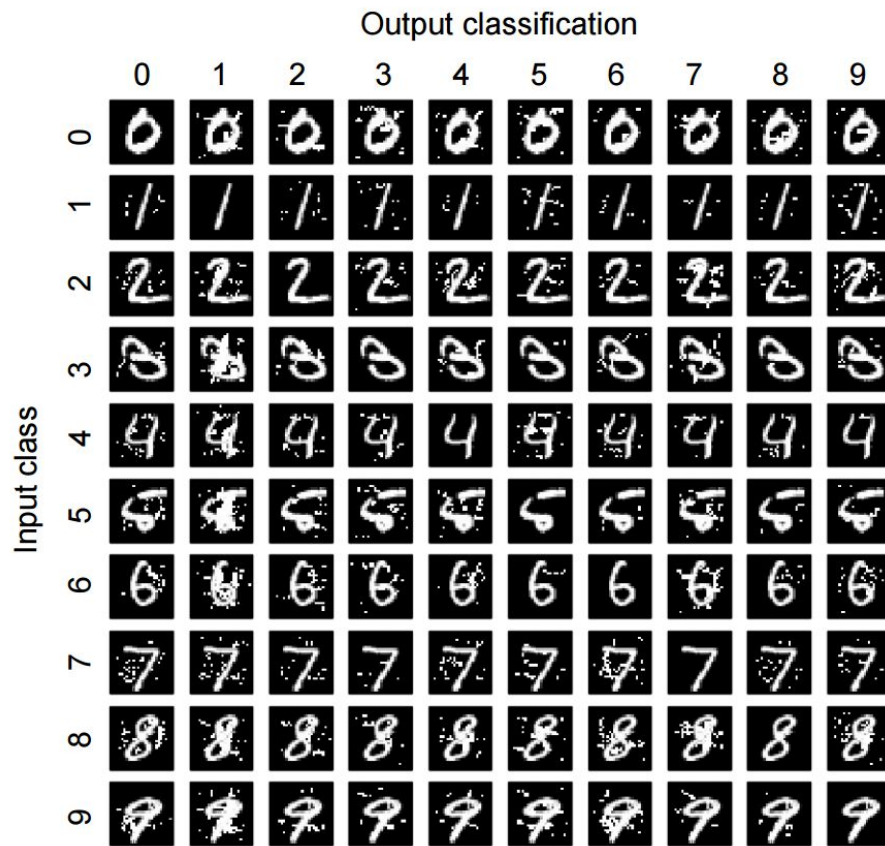


Adversarial examples (white-box attacks)

Jacobian-based Saliency Map Approach (JSMA)



Jacobian-Based Iterative Approach: *source-target misclassification*



Evading a Neural Network Malware Classifier

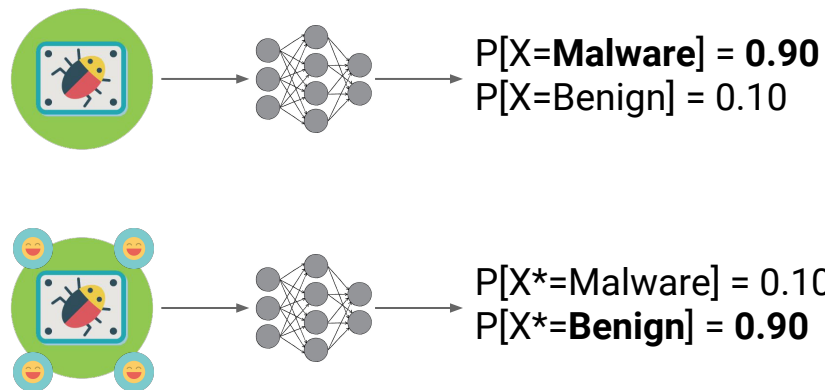
DREBIN dataset of **Android applications**

Add constraints to JSMA approach:

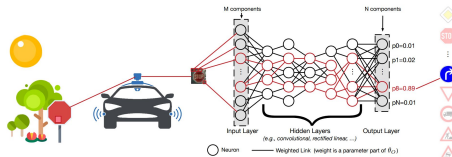
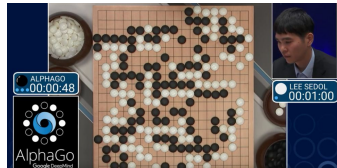
- only *add* features: keep malware behavior
- only features from *manifest*: easy to modify

“Most accurate” neural network

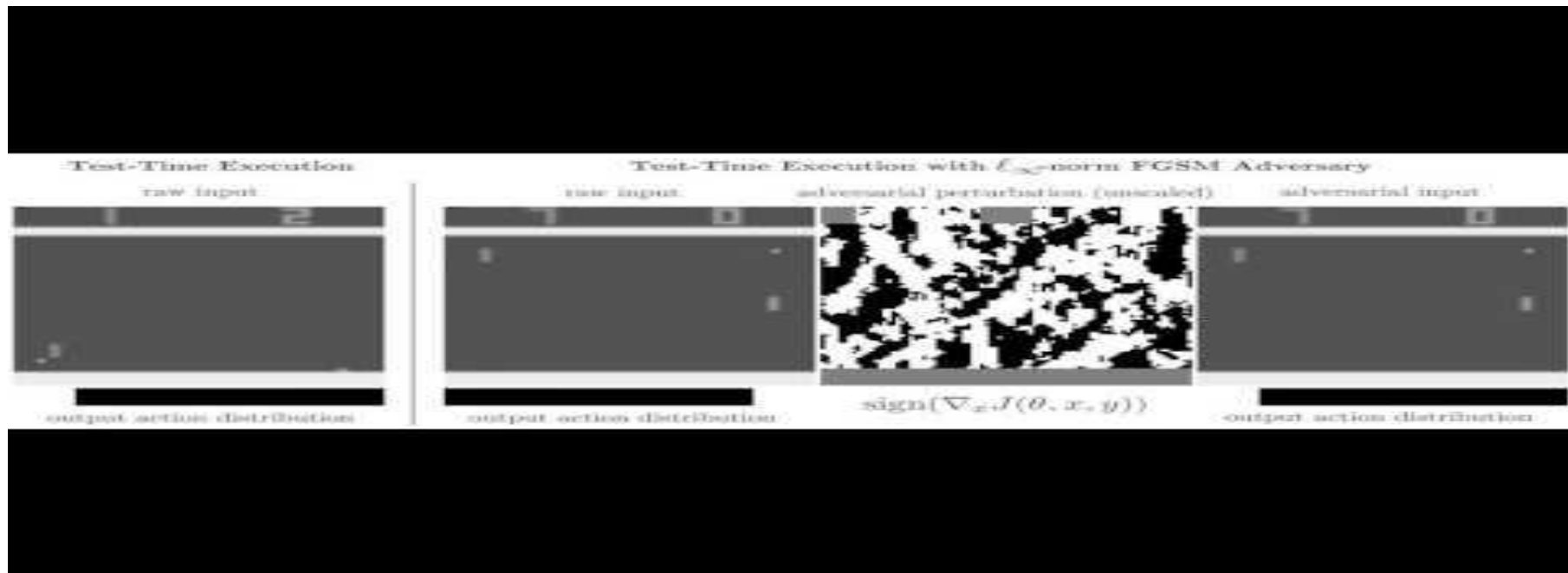
- 98% accuracy, with 9.7% FP and 1.3% FN
- **Evaded with a 63.08% success rate**



Supervised vs. reinforcement learning

	Supervised learning	Reinforcement learning
Model inputs	Observation (e.g., traffic sign, music, email)	Environment & Reward function
Model outputs	Class (e.g., stop/yield, jazz/classical, spam/legitimate)	Action
Training “goal” (i.e., cost/loss)	<i>Minimize class prediction error</i> over pairs of (inputs, outputs)	<i>Maximize reward</i> by exploring the environment and taking actions
Example	 The diagram illustrates a supervised learning neural network for autonomous driving. It shows a car receiving input from a scene with a traffic sign. The input layer has M components, followed by hidden layers, and an output layer with N components. The output layer produces probabilities for different actions like 'stop' or 'yield'.	 The screenshot shows the AlphaGo vs. Lee Sedol Go match. It displays the Go board, AlphaGo's logo, and a timer.

Adversarial attacks on neural network policies



Adversarial examples (black-box attacks)

Threat model of a black-box attack

Adversarial capabilities

~~Training data~~
~~Model architecture~~
~~Model parameters~~
~~Model scores~~



(limited) oracle
access: *labels*

Adversarial goal

Force a ML model remotely accessible through an API to misclassify

Example



Our approach to black-box attacks

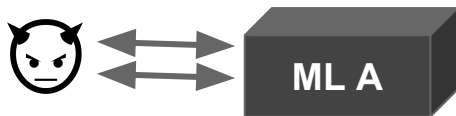
Alleviate lack of knowledge
about model

Alleviate lack of
training data

Adversarial example transferability

Adversarial examples have a **transferability** property:

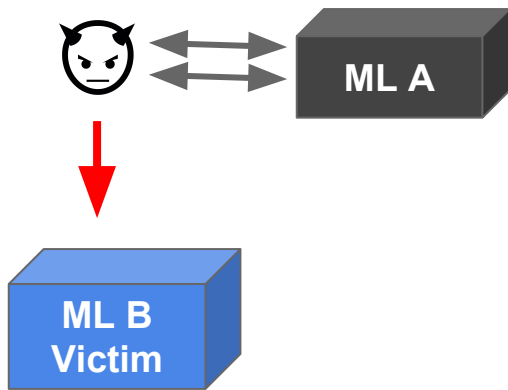
samples crafted to mislead a model A are likely to mislead a model B



Adversarial example transferability

Adversarial examples have a **transferability** property:

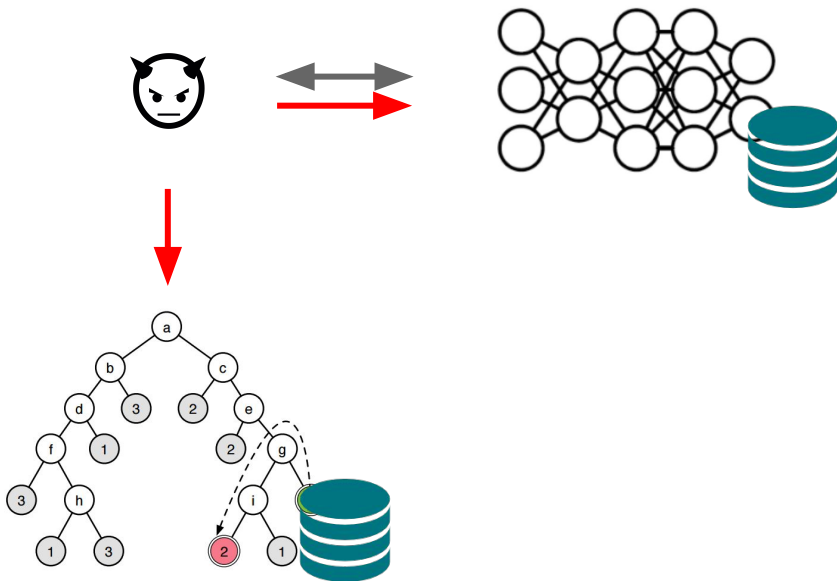
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Adversarial example transferability

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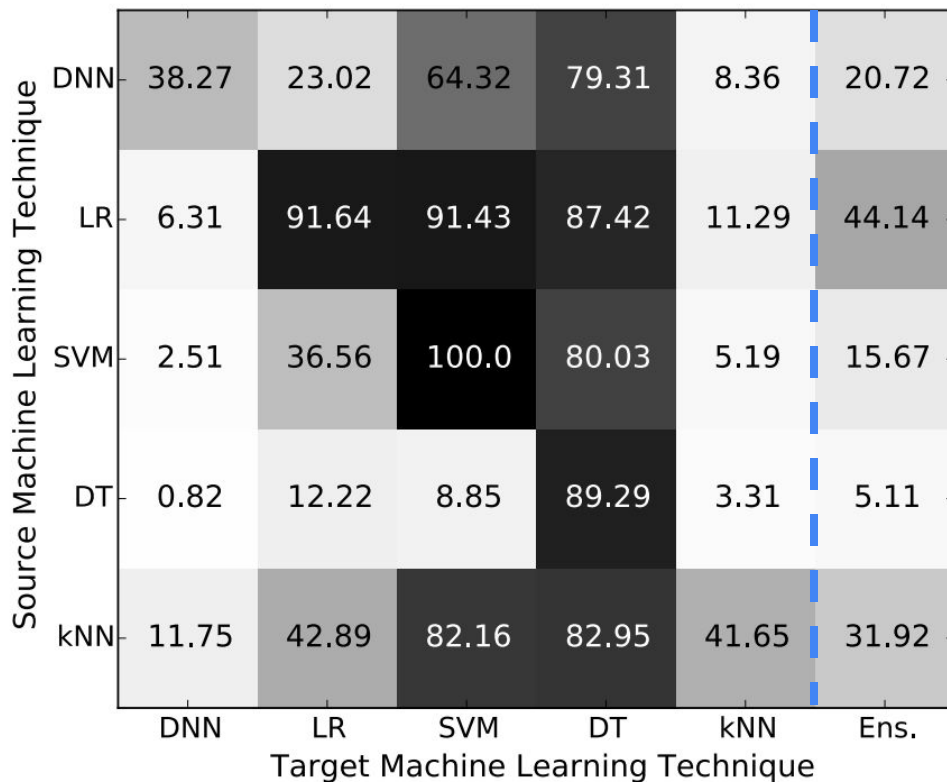
samples crafted to mislead a model A are likely to mislead a model B



Cross-technique transferability

Source Machine Learning Technique	DNN	38.27	23.02	64.32	79.31	8.36
	LR	6.31	91.64	91.43	87.42	11.29
	SVM	2.51	36.56	100.0	80.03	5.19
	DT	0.82	12.22	8.85	89.29	3.31
	kNN	11.75	42.89	82.16	82.95	41.65
		DNN	LR	SVM	DT	kNN
		Target Machine Learning Technique				

Cross-technique transferability



Our approach to black-box attacks

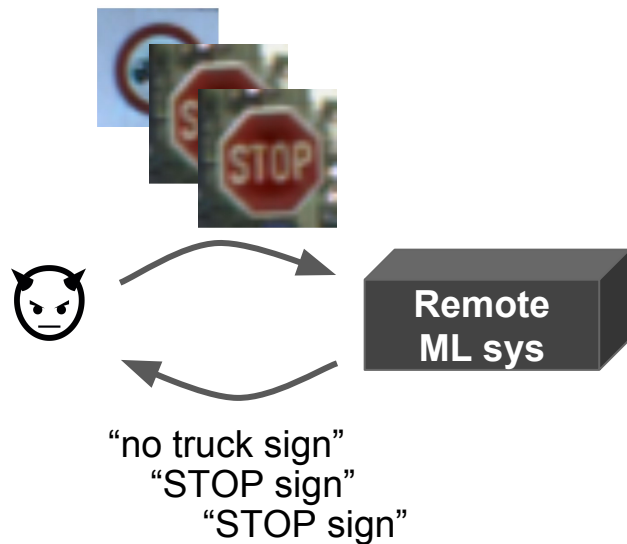
Alleviate lack of knowledge
about model



Adversarial example
transferability from a
substitute model to
target model

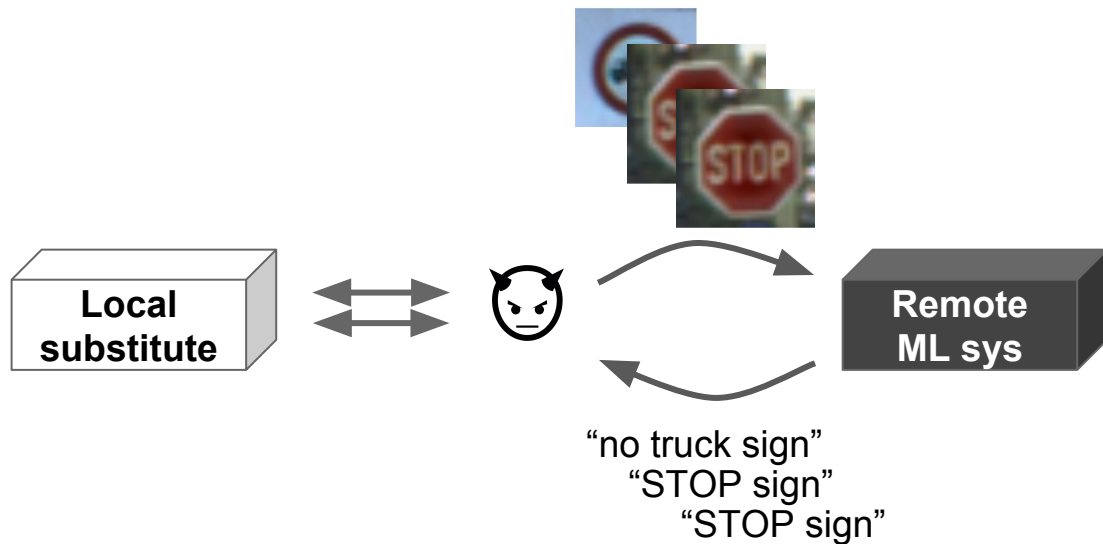
Alleviate lack of
training data

Attacking remotely hosted black-box models



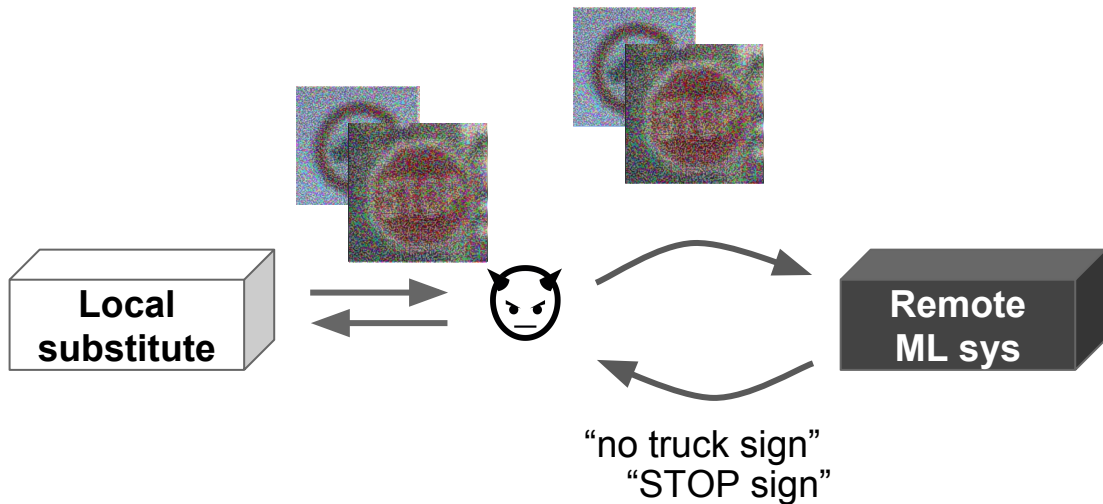
(1) The adversary queries remote ML system for labels on inputs of its choice.

Attacking remotely hosted black-box models



(2) The adversary uses this labeled data to train a local substitute for the remote system.

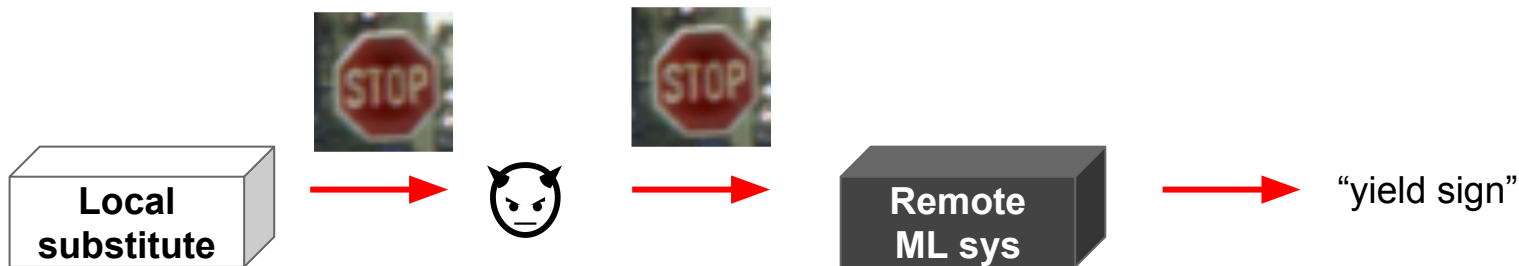
Attacking remotely hosted black-box models



$$S_{\rho+1} = \{\vec{x} + \lambda_{\rho+1} \cdot \text{sgn}(J_F[\tilde{O}(\vec{x})]) : \vec{x} \in S_{\rho}\} \cup S_{\rho}$$

- (3) The adversary selects new synthetic inputs for queries to the remote ML system based on the local substitute's output surface sensitivity to input variations.

Attacking remotely hosted black-box models



- (4) The adversary then uses the local substitute to craft adversarial examples, which are misclassified by the remote ML system because of transferability.

Our approach to black-box attacks

Alleviate lack of knowledge
about model



Adversarial example
transferability from a
substitute model to
target model

+

Alleviate lack of
training data



Synthetic data
generation

Results on real-world remote systems

Remote Platform	ML technique	Number of queries	Adversarial examples misclassified (after querying)
 MetaMind	Deep Learning	6,400	84.24%
 amazon web services™	Logistic Regression	800	96.19%
 Google Cloud Platform	Unknown	2,000	97.72%

All remote classifiers are trained on the MNIST dataset (10 classes, 60,000 training samples)

Benchmarking progress in the adversarial ML community



GitHub repository page for tensorflow / cleverhans. The page shows the repository name, navigation tabs (Code, Issues, Pull requests, Projects, Wiki, Settings, Insights), and a description: "A library for benchmarking vulnerability to adversarial examples". It also displays statistics: 997 commits, 1 branch, 4 releases, 34 contributors, and MIT license.

Below the statistics, there is a table of recent commits and pull requests:

Commit/Pull Request	Author	Message	Time
assets	AlexeyKurakin	add PSD file for logo	2 months ago
cleverhans	goodfeli	Merge pull request #244 from goodfeli/stop_gradient	6 days ago
cleverhans_tutorials	goodfeli	Add the bias to Conv2D computation in tutorial_mnist_tf. The bias was...	6 days ago
examples	goodfeli	Fixing comments	a day ago
tests_tf	goodfeli	move test_attacks_tf into tests_tf folder	6 days ago
tests_th	goodfeli	Update deprecated usage	3 months ago
.gitignore	goodfeli	make accuracy tests run on travis	6 months ago
.travis.yml	goodfeli	tensorflow apparently ignores log level flag	2 months ago
CODE_OF_CONDUCT.rst	goodfeli	update capitalization when using CleverHans as a library name	3 months ago
CONTRIBUTING.md	goodfeli	complete CONTRIBUTING.md file	2 months ago
LICENSE	goodfeli	Add Google Inc. to LICENSE	3 months ago
README.md	goodfeli	avoid implying that contents of master are v2.0.0	7 days ago
requirements.txt	goodfeli	remove theano from requirements and install in Travis only	3 months ago
setup.py	goodfeli	fix typo	2 months ago

The README.md file is displayed below the table, showing the title "CleverHans (latest release: v2.0.0)" and the CleverHans logo.

Growing community

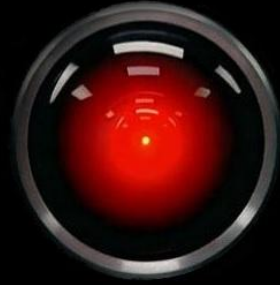
1.3K+ stars

340+ forks

40+ contributors

Adversarial examples represent
worst-case distribution drifts



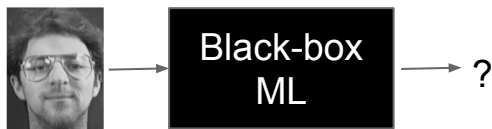


Adversarial examples are a *tangible*
instance of hypothetical AI safety problems

Part II

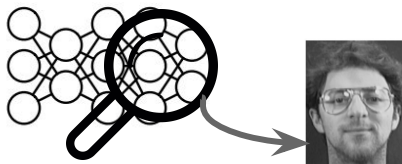
Privacy in machine learning

Types of adversaries and our threat model



Model querying (**black-box adversary**)

Shokri et al. (2016) *Membership Inference Attacks against ML Models*
Fredrikson et al. (2015) *Model Inversion Attacks*



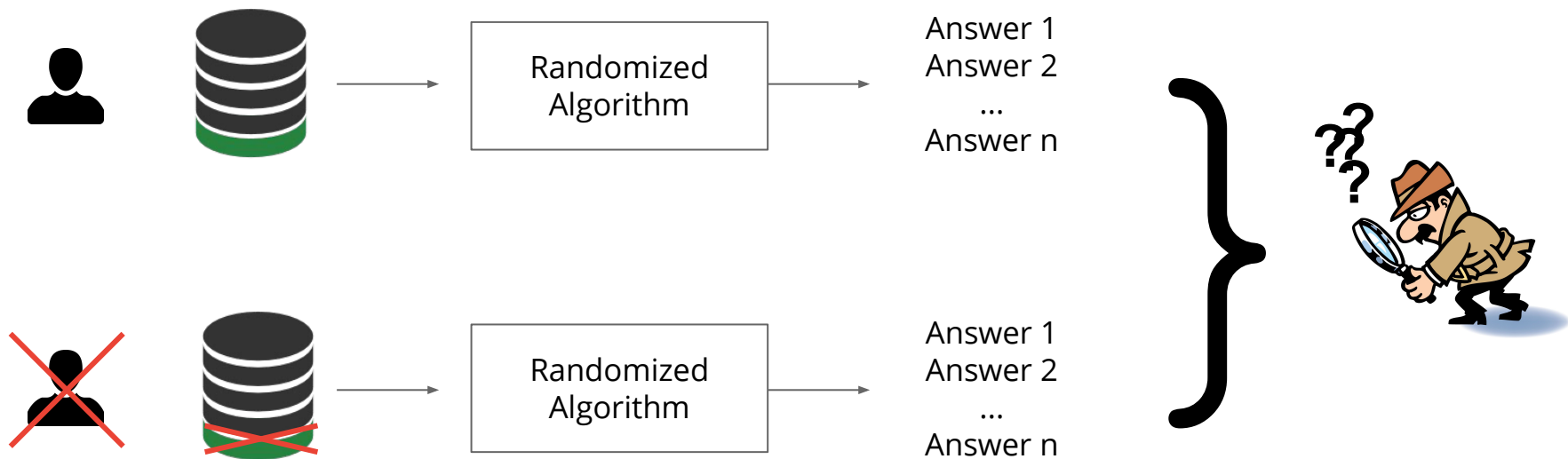
Model inspection (**white-box adversary**)

Zhang et al. (2017) *Understanding DL requires rethinking generalization*

In our work, the threat model assumes:

- Adversary can make a potentially unbounded number of queries
- Adversary has access to model internals

A definition of privacy



Our design goals

Problem Preserve **privacy of training data** when learning **classifiers**

Goals **Differential privacy** protection guarantees
 Intuitive privacy protection guarantees
 Generic* (independent of learning algorithm)

*This is a key distinction from previous work, such as

Pathak et al. (2011) *Privacy preserving probabilistic inference with hidden markov models*

Jagannathan et al. (2013) *A semi-supervised learning approach to differential privacy*

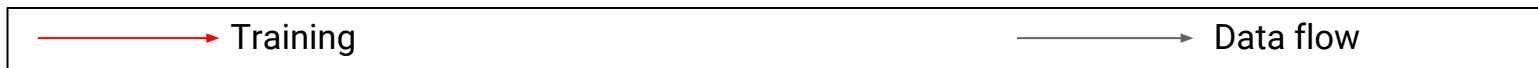
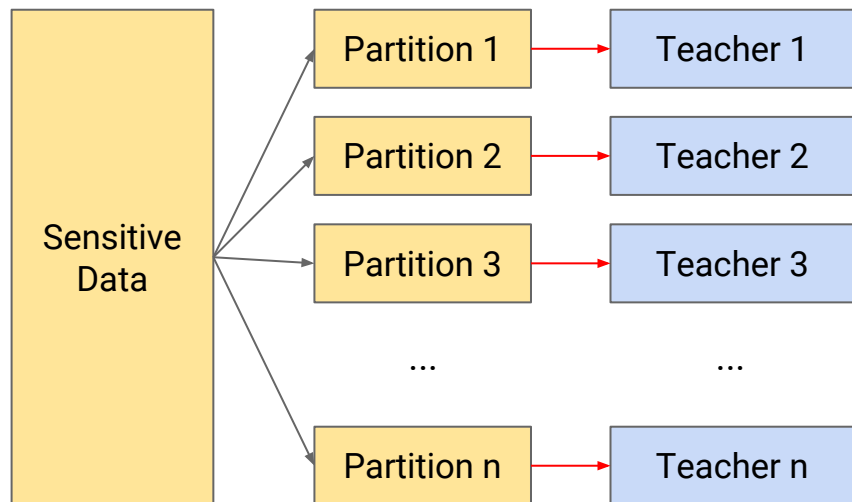
Shokri et al. (2015) *Privacy-preserving Deep Learning*

Abadi et al. (2016) *Deep Learning with Differential Privacy*

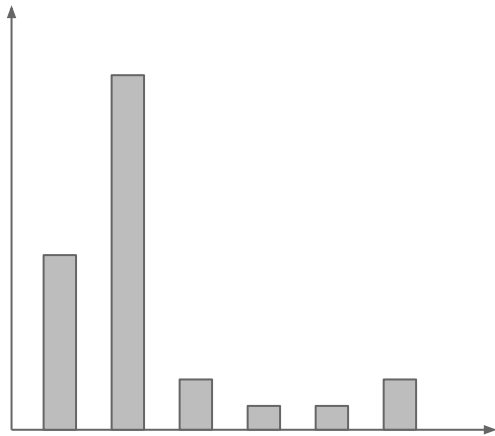
Hamm et al. (2016) *Learning privately from multiparty data*

The PATE approach

Teacher ensemble

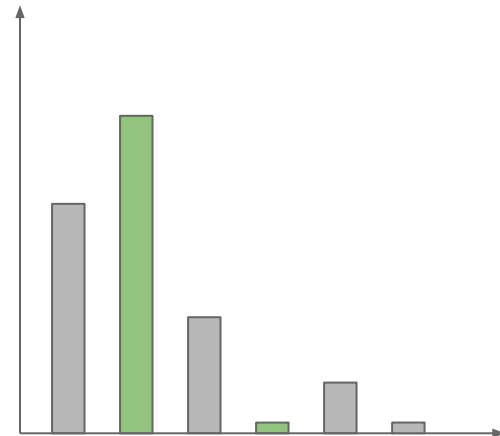


Aggregation



Count votes

$$n_j(\vec{x}) = |\{i : i \in 1..n, f_i(\vec{x}) = j\}|$$

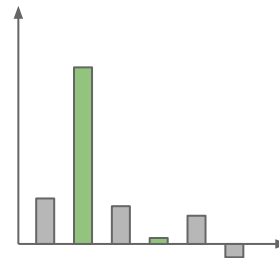


Take maximum

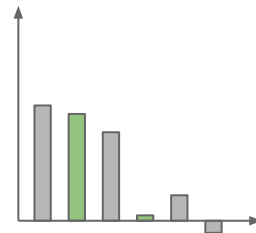
$$f(x) = \arg \max_j \left\{ n_j(\vec{x}) \right\}$$

Intuitive privacy analysis

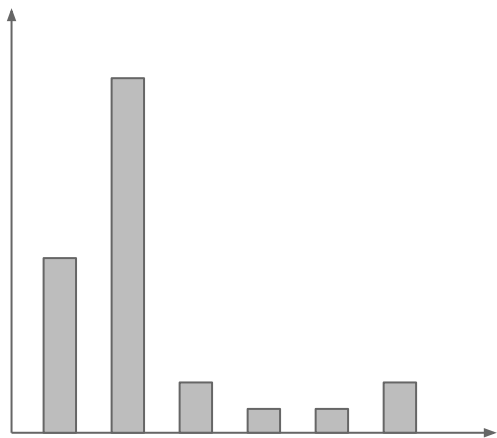
If most teachers agree on the label, it does not depend on specific partitions, so the privacy cost is small.



If two classes have close vote counts, the disagreement may reveal private information.

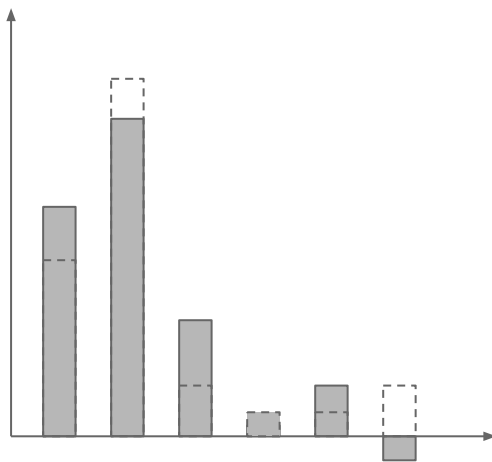


Noisy aggregation



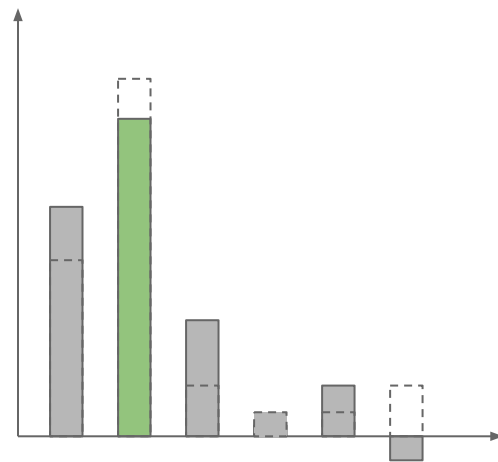
Count votes

$$n_j(\vec{x}) = |\{i : i \in 1..n, f_i(\vec{x}) = j\}|$$



Add Laplacian noise

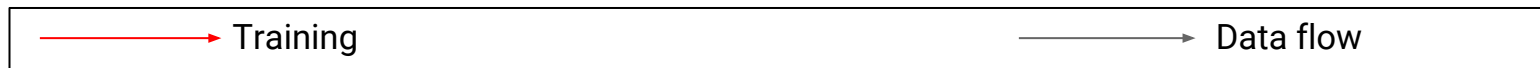
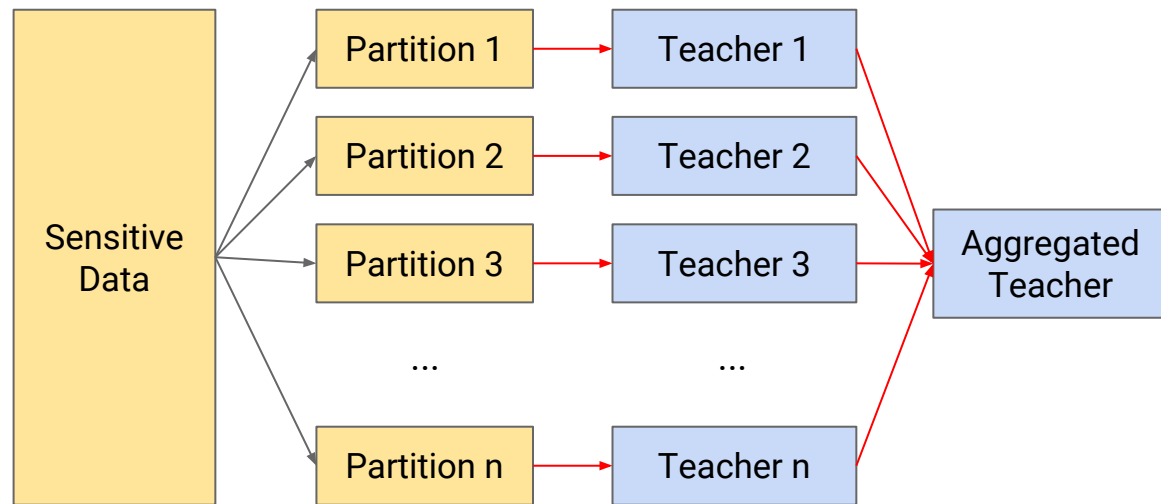
$$Lap\left(\frac{1}{\varepsilon}\right)$$



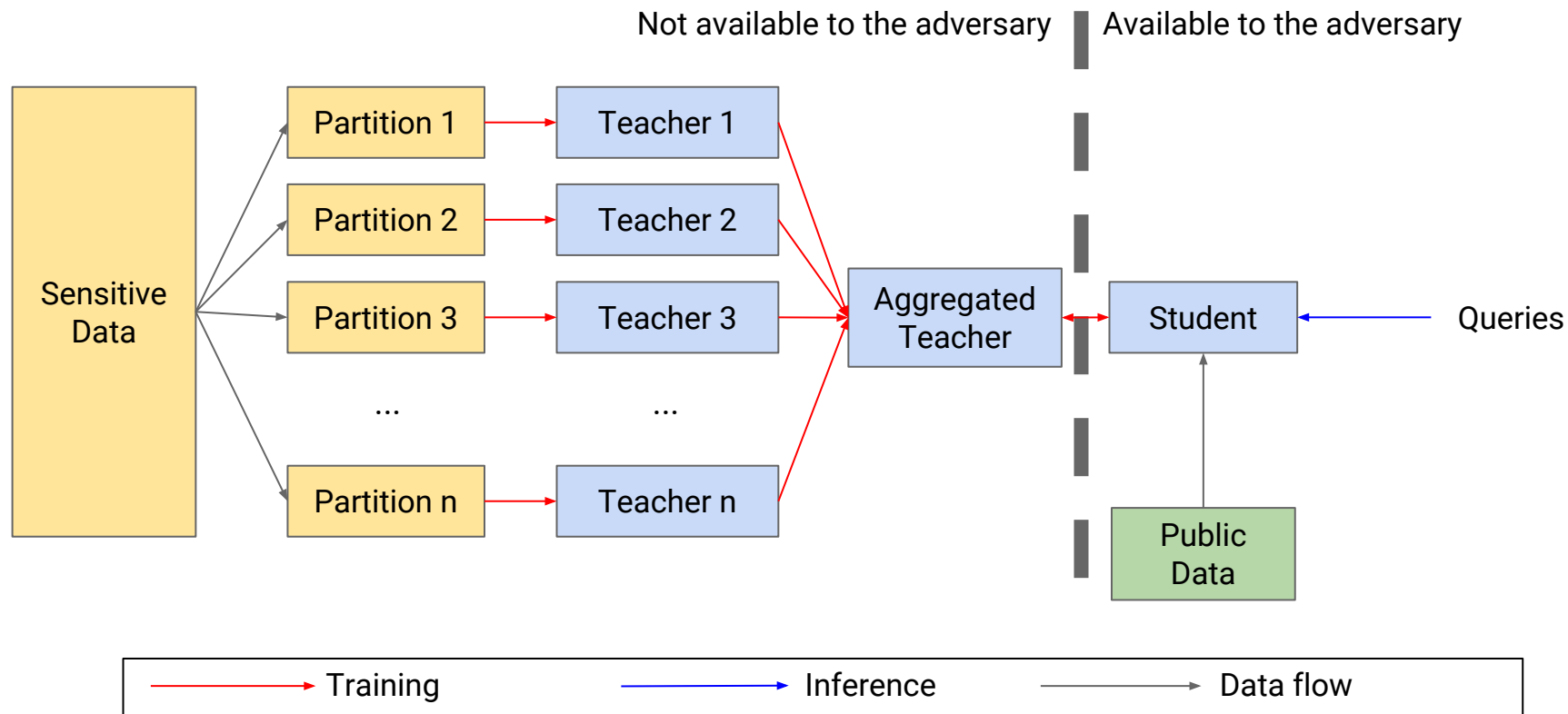
Take maximum

$$f(x) = \arg \max_j \left\{ n_j(\vec{x}) + Lap\left(\frac{1}{\varepsilon}\right) \right\}$$

Teacher ensemble



Student training

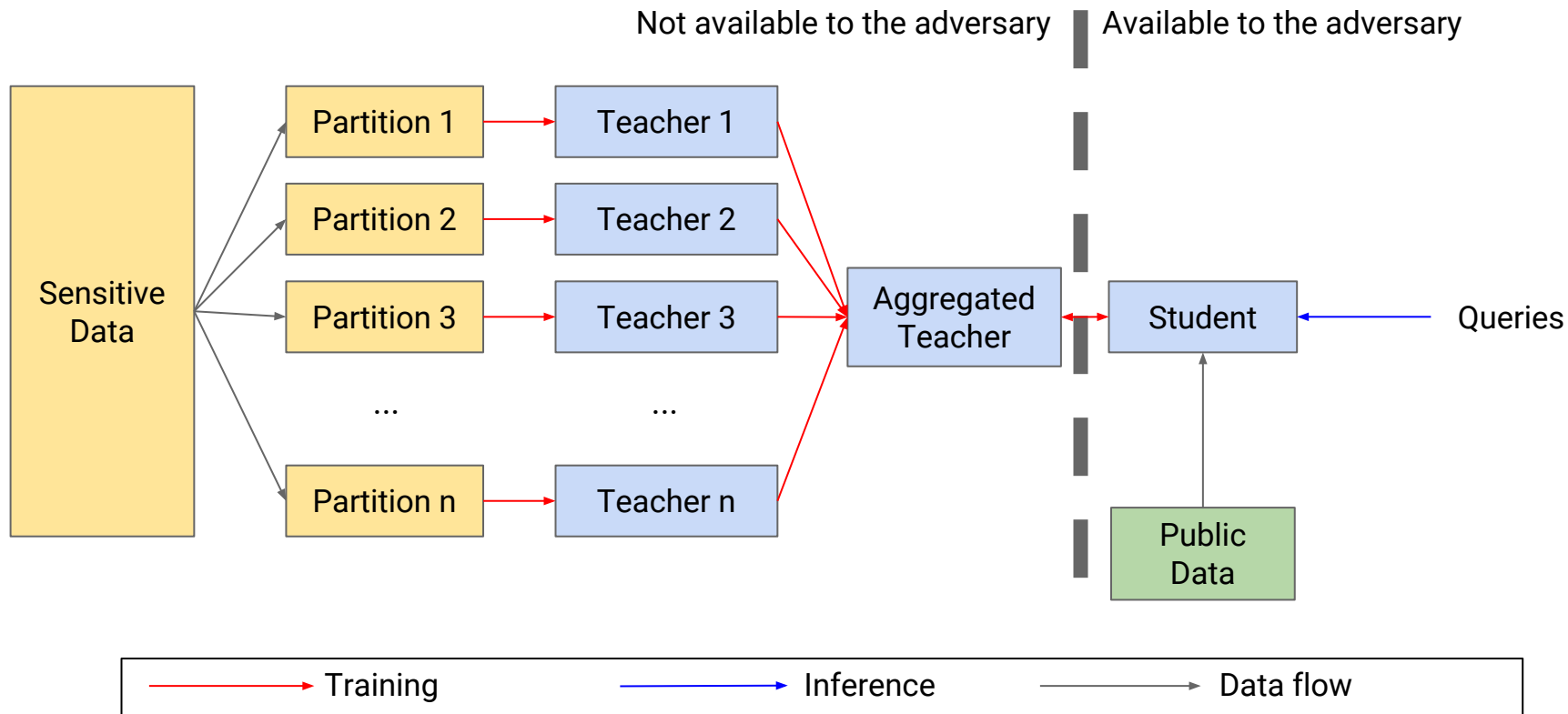


Why train an additional “student” model?

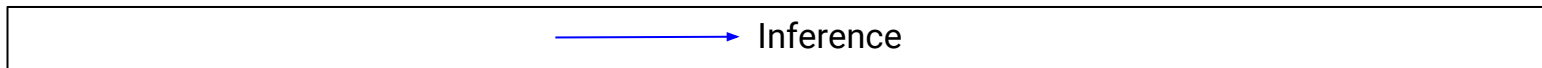
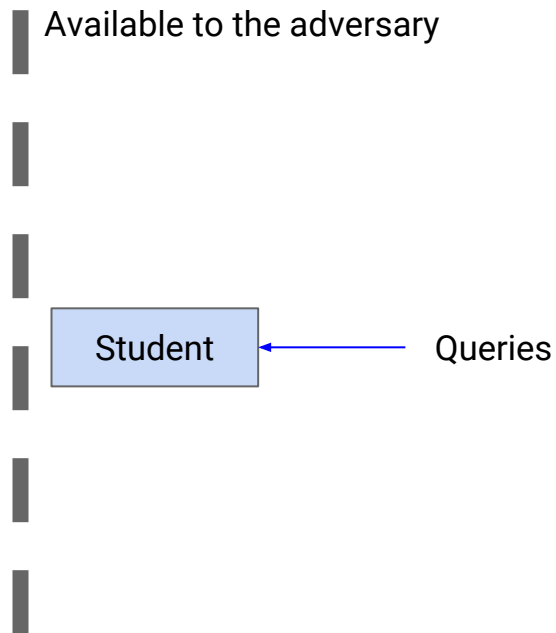
The aggregated teacher violates our threat model:

- 1 Each prediction increases total privacy loss.
Privacy budgets create a tension between the accuracy and number of predictions.
- 2 Inspection of internals may reveal private data.
Privacy guarantees should hold in the face of white-box adversaries.

Student training



Deployment



Differential privacy analysis

Differential privacy:

A randomized algorithm M satisfies (ϵ, δ) differential privacy if for all pairs of neighbouring datasets (d, d') , for all subsets S of outputs:

$$\Pr[M(d) \in S] \leq e^\epsilon \Pr[M(d') \in S] + \delta$$

Application of the **Moments Accountant** technique (Abadi et al, 2016)

Strong **quorum** $\Rightarrow$ Small privacy cost

Bound is **data-dependent**: computed using the empirical quorum

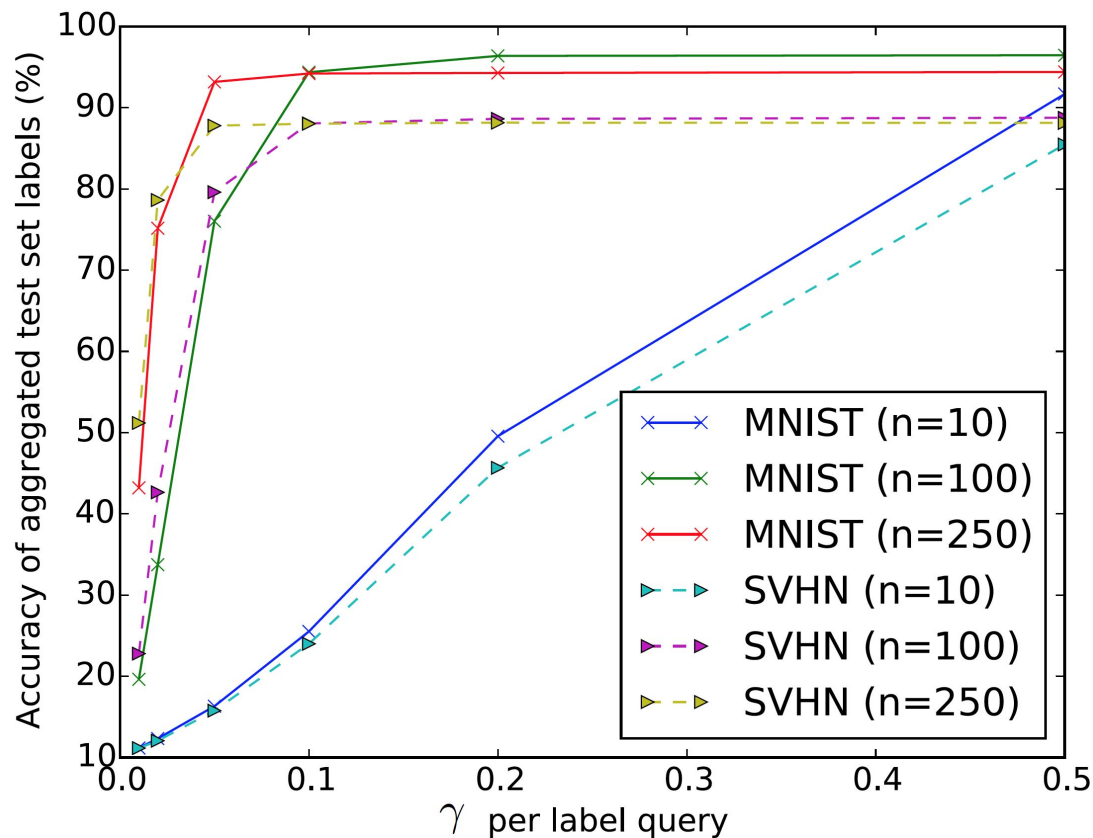
Experimental results

Experimental setup

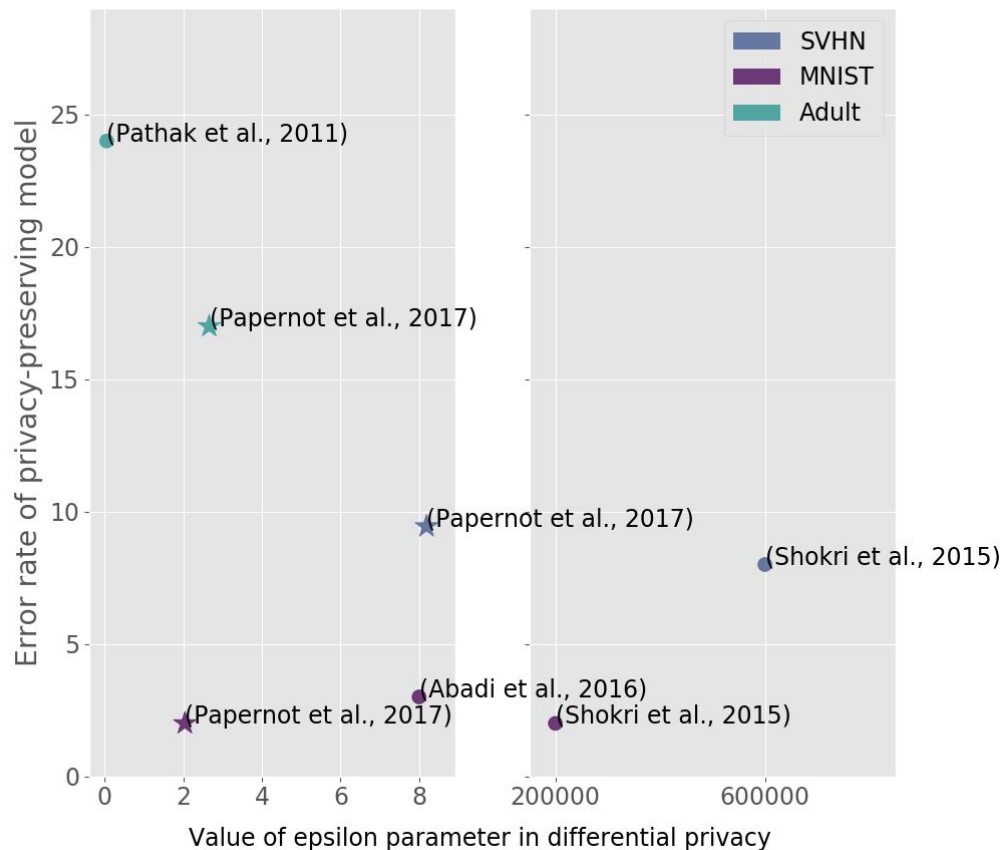
Dataset	Teacher Model	Student Model
MNIST	Convolutional Neural Network	Generative Adversarial Networks
SVHN	Convolutional Neural Network	Generative Adversarial Networks
UCI Adult	Random Forest	Random Forest
UCI Diabetes	Random Forest	Random Forest

 /  TensorFlow™ / **models** / tree/master/differential_privacy/multiple_teachers

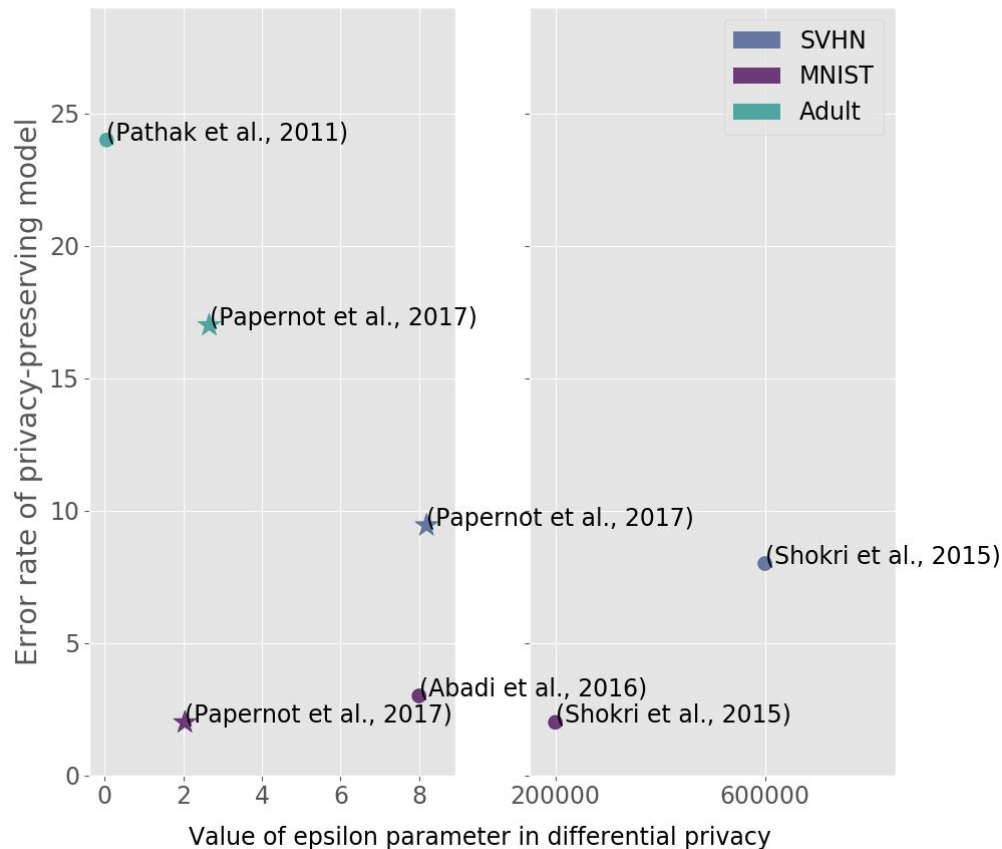
Aggregated teacher accuracy



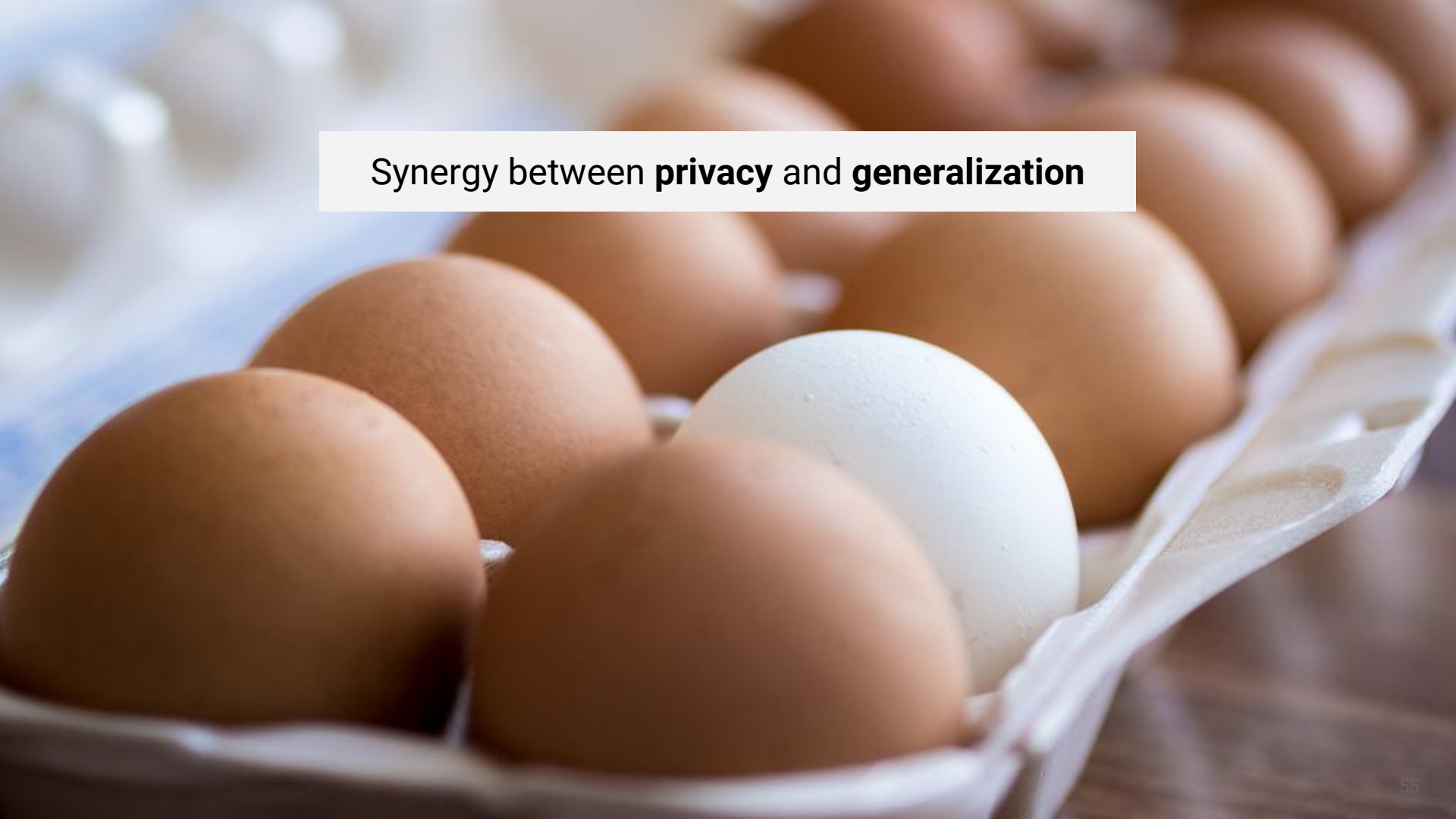
Trade-off between student accuracy and privacy



Trade-off between student accuracy and privacy



UCI Diabetes	
ϵ	1.44
δ	10^{-5}
Non-private baseline	93.81%
Student accuracy	93.94%

A close-up photograph of a white egg carton filled with brown eggs. One white egg is prominently placed in the center, standing out from the surrounding brown eggs. The background is softly blurred, showing more of the same scene.

Synergy between **privacy** and **generalization**

Some online resources:

Blog on S&P in ML (joint work w/ Ian Goodfellow)

ML course

DL course

www.cleverhans.io

<https://coursera.org/learn/machine-learning>

<https://coursera.org/learn/neural-networks>

Assigned reading and more in-depth technical survey paper:

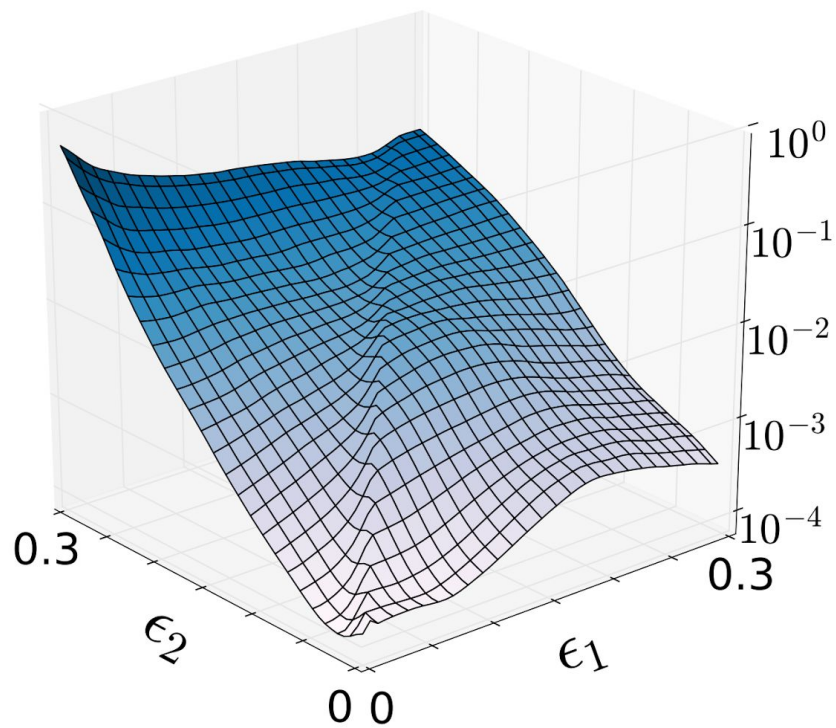
Machine Learning in Adversarial Settings

Patrick McDaniel, Nicolas Papernot, Z. Berkay Celik

Towards the Science of Security and Privacy in Machine Learning

Nicolas Papernot, Patrick McDaniel, Arunesh Sinha, and Michael Wellman

Gradient masking



Gradient masking

